

Question	Answer	Mark	Mark scheme	Additional guidance
21	Proof	M1	for $\overrightarrow{DQ} = \frac{1}{2}(\mathbf{b} - \mathbf{a})$ oe or $\overrightarrow{EQ} = \frac{1}{2}(\mathbf{a} - \mathbf{b})$ oe	Vectors could be written on the diagram
		M1	for $\overrightarrow{PQ} = \frac{1}{2}\mathbf{a} + \overrightarrow{DQ}$ or $\frac{1}{2}\mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a})$ oe or $\overrightarrow{PQ} = -\frac{1}{2}\mathbf{a} + \mathbf{b} + \overrightarrow{EQ}$ or $-\frac{1}{2}\mathbf{a} + \mathbf{b} + \frac{1}{2}(\mathbf{a} - \mathbf{b})$ oe	
		B1	for $\overrightarrow{PQ} = \frac{1}{2}\mathbf{b}$	
		C1	for complete proof with statement, eg $FE = 2PQ$ or FE is a multiple of PQ or $\mathbf{b} = 2(\frac{1}{2}\mathbf{b})$	